# Hybrid Intrusion Detection System Using Supervised and Unsupervised Learning Algorithm

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Github Link- https://github.com/networkenthusiast/AI-Network-Intrusion-Detection.git

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**Karthik Mekala** significantly contributed to data analysis, preprocessing, and the implementation of unsupervised learning techniques using K-Means clustering. His work included data balancing, feature scaling, dimensionality reduction using PCA, and extraction of clustering-based features. He also contributed to model evaluation, comparative analysis, and preparation of technical presentation and documentation.

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## 1. INTRODUCTION

In recent years, cyber security threats have increased significantly, particularly for systems handling sensitive data. Cyber-attacks now occur at an alarming frequency, leading to large-scale data breaches, infrastructure disruptions, and financial losses. For instance, incidents such as massive data leaks and exposed records highlight the growing vulnerability of digital systems. Global cybercrime costs are projected to rise substantially, emphasizing the urgency for robust security mechanisms.

Traditional security measures like firewalls are limited in their ability to detect modern threats. They primarily rely on predefined rules and signatures, making them ineffective against evolving and unknown attacks, encrypted traffic, and internal misuse.

To overcome these limitations, machine learning techniques have been integrated into cybersecurity. These approaches enable intelligent detection of anomalous behaviour, adaptability to new attack patterns, and efficient analysis of large-scale network data, thereby enhancing overall intrusion detection capabilities.

## 2. LITERATURE REVIEW

Research on the CICIDS2017 dataset shows that supervised models like Random Forest and XGBoost achieve high accuracy (>99%) [1], [2], [4], but often face overfitting and limited generalization [2]. PCA is commonly used for dimensionality reduction, improving efficiency but sometimes losing important information [6], [7].Hybrid methods combining clustering (e.g., K-Means) with classifiers enhance performance [3], though clustering alone is insufficient. Ensemble models like XGBoost improve accuracy but rely heavily on labeled data and struggle with unknown attacks [4], while stacking methods lack integration with unsupervised learning [5].Overall, limitations include dependence on supervised learning, class imbalance, and poor detection of novel attacks. The proposed hybrid model addresses these by combining Random Forest, PCA + K-Means, and XGBoost for improved generalization and detection.

# 3. DATA SUMMARY

## 3.1 Dataset Description

- [1], The study uses the **CICIDS2017 dataset** from Kaggle, containing labelled network traffic with multiple attack types and 78 flow-based features suitable for machine learning.
- The dataset was generated in a realistic testbed environment at the Canadian Institute for Cybersecurity, capturing both benign and malicious traffic.
- Data collection spans five days, including one fully benign day and multiple days with attacks such as DoS, DDoS, brute force, web attacks, botnet, and infiltration.
- The dataset reflects modern network behaviour and is widely used for intrusion detection research as a replacement for older datasets.
- Some limitations include class imbalance, label noise, and redundant or irrelevant flows, which require preprocessing.

## 3.2 Data Pre-Processing

- Cleaning negative values – Negative or invalid values were handled by removing a few records and replacing others with zero to avoid excessive data loss.
- Column consistency check – Ensured all CSV files had identical column structure before merging to prevent data misalignment.
- Removal of infinite values – Infinite values arising from division-based features were identified and removed to ensure stable model training.
- Duplicate column removal – Redundant or duplicate columns were eliminated to improve data quality and reduce dimensionality.

### 3.3 Major challenges in Network Distribution Labels

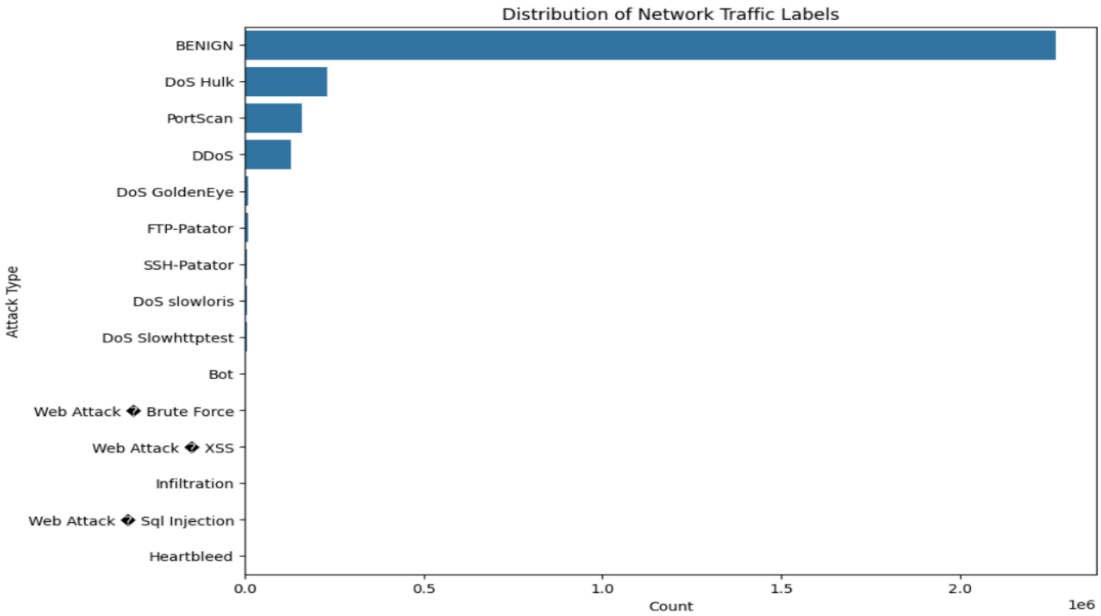


*Fig 3.3: Distribution of network table labels*

The figure shows a highly imbalanced dataset, where BENIGN traffic overwhelmingly dominates. Among attacks, DoS Hulk, Port Scan, and DDoS appear more frequently, while others like Heartbleed, SQL Injection, Infiltration, and Web Attacks have very few samples.

•This imbalance can bias the model toward predicting the BENIGN class.

• Rare attack types may not be effectively learned by the model.

### 3.4 Correlation Analysis of Numerical Features

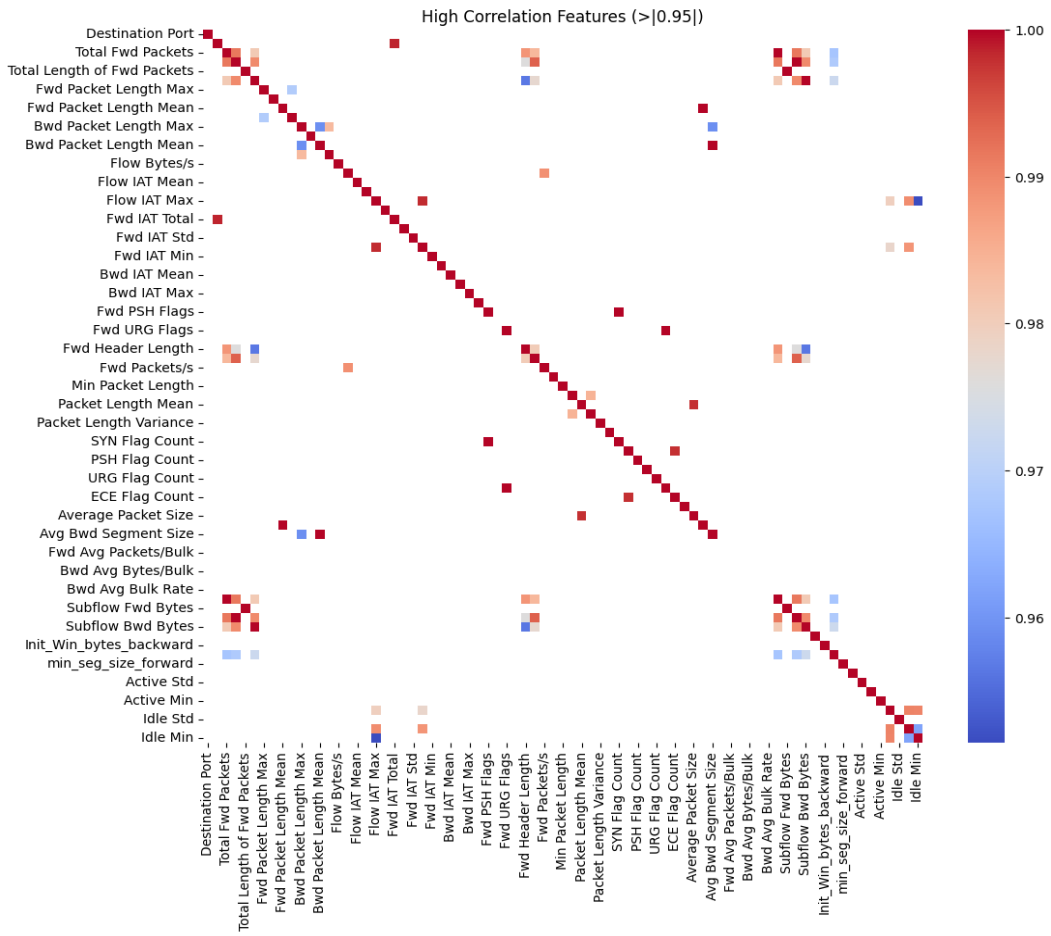


*Fig 3.5: Correlation Features*

The analysis identified several highly correlated and redundant features. Removing them reduced dimensionality, improved training efficiency, minimized overfitting, and enhanced interpretability.

Overall, correlation-based feature selection which we will see in the following section , streamlined the dataset while retaining important information, resulting in a more efficient and robust feature set for model training.

### 3.5 Binary Label Encoding

The dataset is converted into a binary classification problem by encoding BENIGN traffic as 0 and all attack types as 1. Labels are first standardized (trimmed and uppercased) for consistency. This simplifies the task and aligns with practical intrusion detection systems.

### 3.6 Train–Test Split and Class Imbalance Handling

The dataset is imbalanced, with benign traffic dominating attack instances. Stratified sampling is used to split the data while preserving class distribution.

Under sampling is applied only to the training set to balance classes and avoid data leakage. This ensures improved detection of attack traffic while maintaining realistic evaluation conditions.

## 4. METHODOLOGY

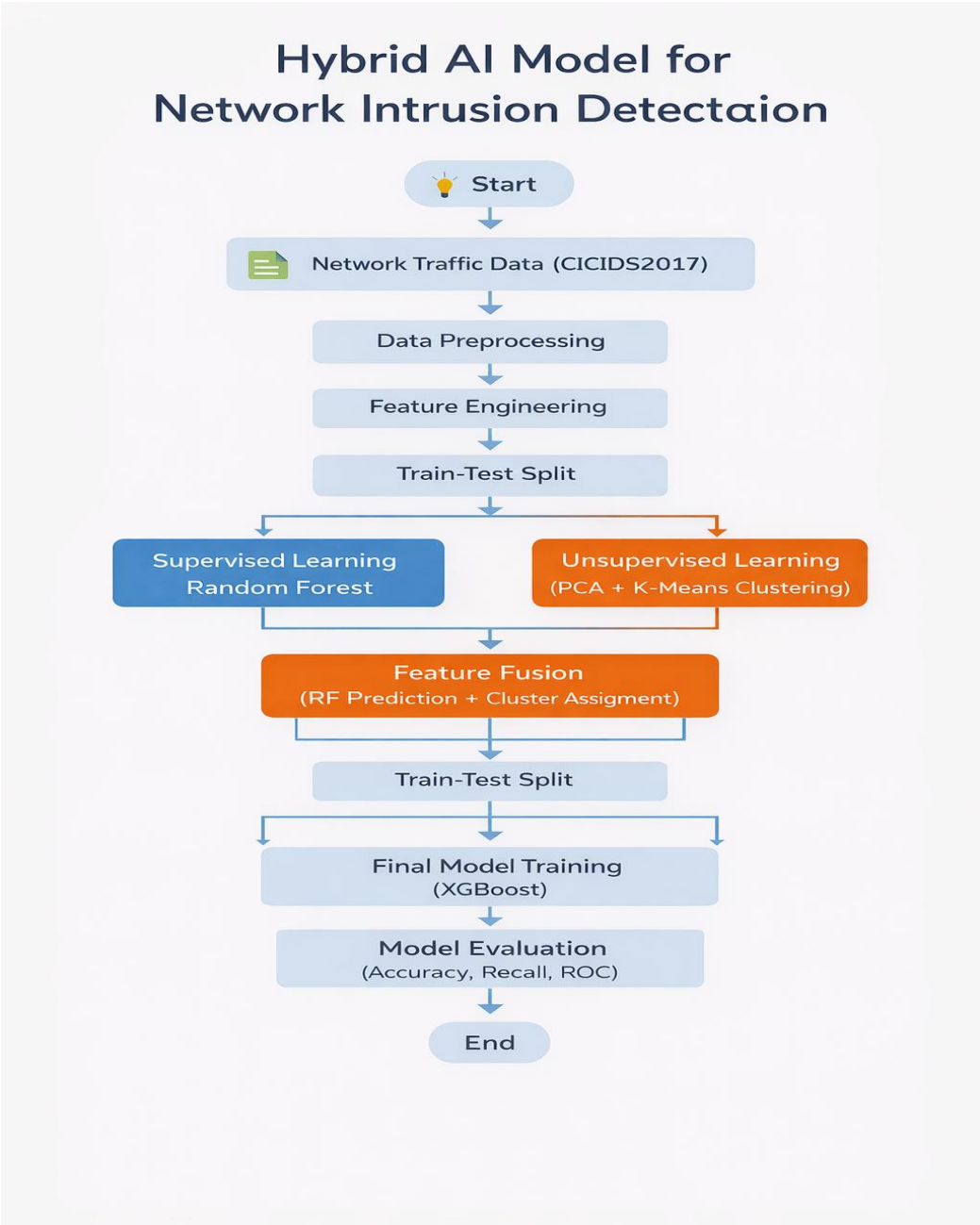


*Fig 4: System Architecture of the Proposed Model*

The model follows a hybrid approach combining supervised and unsupervised learning. A Random Forest generates initial predictions, while PCA and K-Means capture underlying data patterns.

These outputs are fused to create an enhanced feature set, which is used to train a final XGBoost model for improved intrusion detection performance.

## 4.1 MODEL 1: SUPERVISED LEARNING APPROACH(RANDOM FOREST)

### 4.1.1 SYSTEM OVERVIEW

Model 1 uses a supervised learning approach to classify network traffic as normal or malicious using labelled data. It follows a pipeline of preprocessing, feature engineering, training, and evaluation. A Random Forest classifier is applied on the CICIDS2017 dataset, with performance improved through feature selection, hyperparameter tuning, and evaluation.

#### 4.1.1.1 Data Preparation

The dataset was split into features (X) and target (y), where *Traffic_Label* represents the binary class. A balanced training dataset was used to handle class imbalance and ensure fair learning across both classes.

#### 4.1.1.2 Model Selection

A Random Forest classifier was chosen for its ability to handle high-dimensional data, resist overfitting, and provide feature importance. It was configured with random_state=42 for reproducibility and n_jobs=-1 for parallel processing.

#### 4.1.1.3 Hyperparameter Tuning

Grid Search with 5-fold Stratified Cross-Validation identified optimal parameters: max_depth=20, min_samples_leaf=2, min_samples_split=5, and n_estimators=100. The top 15 features were selected based on random forest feature importance to improve efficiency and interpretability.

#### 4.1.1.4 Model Retraining

The model was retrained using the selected top features, with X_train_top and X_test_top as the training and testing sets. This enables the model to focus on the most relevant information.

### 4.1.2 MODEL EVALUATION AND RESULTS

#### 4.1.2.1 Classification Report

The model was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. The results indicate strong overall performance, with an accuracy of approximately 99% and high precision and recall across both classes.

```
Classification Report:
              precision    recall  f1-score   support

           0       1.00      0.99      1.00    453718
           1       0.98      1.00      0.99    111273

    accuracy                           0.99    564991
   macro avg       0.99      1.00      0.99    564991
weighted avg       0.99      0.99      0.99    564991
```

*Fig 4.1.2.1: Classification Report (Supervised Model-Random Forest)*

The model achieves strong and balanced performance with F1-scores of 0.99.

- Benign (Class 0): High precision (1.00) and recall (0.99) indicate accurate classification with minimal false positives.
- Attack (Class 1): Perfect recall (1.00) ensures almost all attacks are detected, with high precision (0.98).

Interpretation:

The model effectively distinguishes between normal and malicious traffic, maintaining a strong precision–recall balance, with high attack detection and improved efficiency through feature selection.

#### 4.1.2.2 CONFUSION MATRIX ANALYSIS

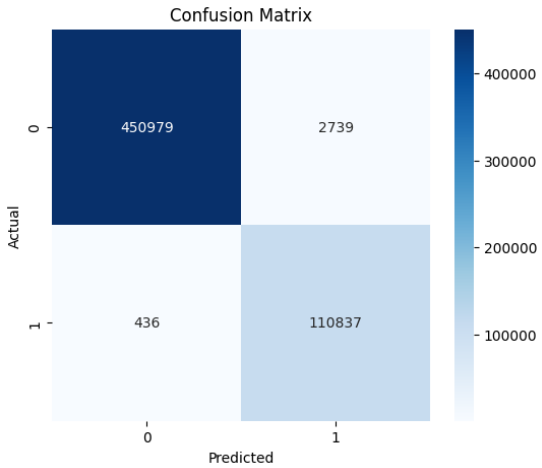


*Fig 4.1.2.2: Confusion Matrix (Supervised Model-Random Forest)*

The Random Forest model shows strong performance with high true negatives (450,979) and true positives (110,837), indicating effective detection of both benign and malicious traffic. False positives (2,739) are low, while false negatives (436) are minimal, ensuring most attacks are correctly identified.Overall, the model demonstrates reliable intrusion detection performance, with low misclassification rates and strong emphasis on detecting malicious activity.

#### 4.1.2.3 ROC CURVE ANALYSIS

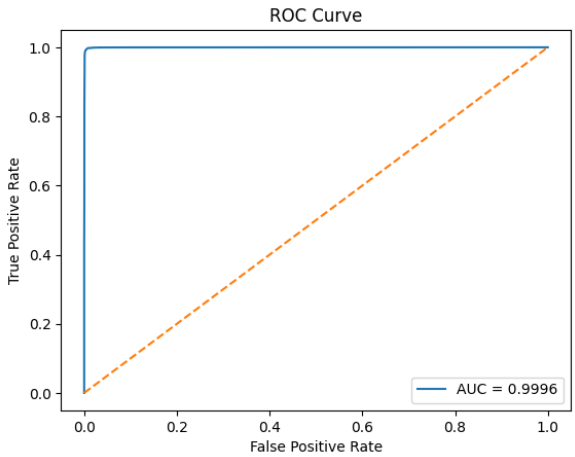


*Fig 4.1.2.3: ROC Curve (Supervised Model--Random Forest)*

The ROC curve shows excellent model performance, with an AUC of 0.9996, indicating near-perfect class separation. The curve's proximity to the top-left corner reflects a high true positive rate and low

false positive rate. This confirms the model's strong ability to distinguish between benign and malicious traffic, making it highly effective for intrusion detection, with further validation recommended for generalization.

#### 4.1.2.4 FEATURE IMPORTANCE ANALYSIS

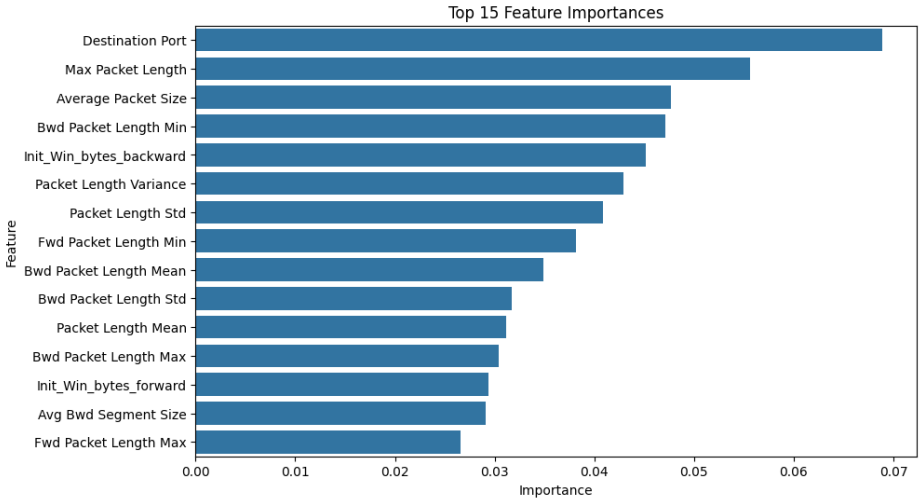


*Fig 4.1.2.4: Top 15 Feature Importance (Supervised Model--Random Forest)*

Random Forest feature importance identifies key features such as Destination Port, packet length metrics, and forward/backward traffic characteristics, along with TCP window size features for connection-level insights. Overall, traffic patterns are crucial for attack detection. Selecting top features improves efficiency, reduces dimensionality, and enhances interpretability while maintaining performance.

#### 4.1.2.5 LEARNING CURVE ANALYSIS

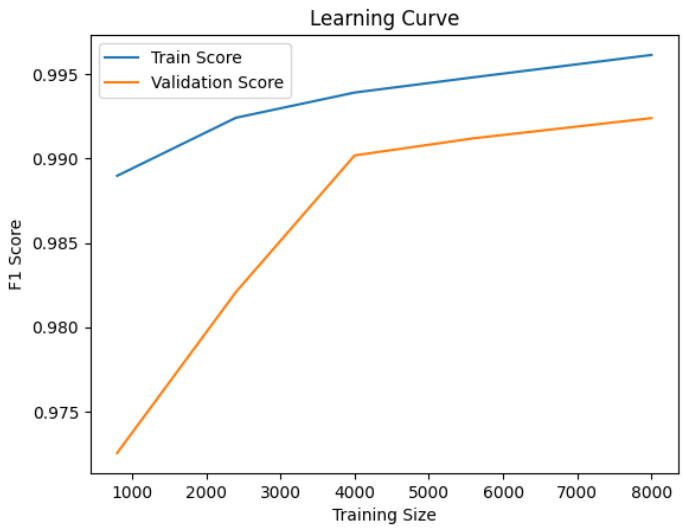


*Fig 4.1.2.5: Learning Curve (Supervised Model--Random Forest)*

The model shows strong performance, with training F1-score increasing from **0.989 to 0.996** and validation F1-score from **0.972 to 0.992** as data size grows. The small gap (~0.003–0.004) indicates minimal overfitting and good generalization, showing a stable and well-balanced model.

## 4.2 MODEL 2 : UNSUPERVISED LEARNING APPROACH(K-Means Clustering)

### 4.2.1 SYSTEM OVERVIEW:

K-Means clustering is used to extract hidden patterns and generate new features to enhance the final supervised model.

#### 4.2.1.1 Balancing Sample:

A balanced dataset of 50,000 benign and 50,000 attack samples is created to avoid bias in clustering.

#### 4.2.1.2 Data Preparation:

The same test set from Model 1 is reused for consistency.

#### 4.2.1.3 Feature Scaling:

StandardScaler is applied to normalize features, ensuring fair distance-based clustering.

#### 4.2.1.4 Dimensionality Reduction using PCA

Principal Component Analysis (PCA) is applied to reduce the dimensionality of the dataset, transforming the original 77 features into 20 principal components. This dimensionality reduction helps eliminate noise and redundant information, enhances clustering performance, and reduces computational complexity.

#### 4.2.1.5 K-Means Clustering

MiniBatch K-Means is applied to perform clustering with a configuration of two clusters (K = 2), a batch size of 5,000, and 10 initialization runs. This approach is selected due to its computational efficiency on large datasets, as it processes data in mini-batches while providing a close approximation to standard K-Means clustering.

#### 4.2.1.6 Cluster Interpretation

Since K-Means is unsupervised, clusters do not directly map to class labels. Mean label values are used with a 0.5 threshold to assign meaning.Cluster 0 (mean ≈ 0.40) is mapped to benign, while Cluster 1 (mean ≈ 0.88) is mapped to attack, indicating clear separation between traffic types.

#### 4.2.1.7 Visualization of K-Means Clustering in PCA Space

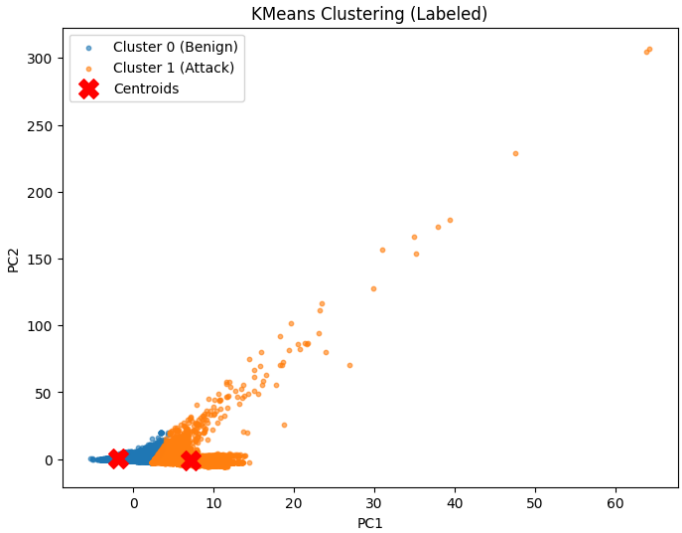


*Fig 4.2.1.7: K-Means Clustering in PCA Space*

The plot shows clear separation between clusters using PC1 and PC2. Benign traffic is concentrated near the origin, while attack traffic is more dispersed. Centroids lie at cluster centres, indicating distinct groupings. Despite slight overlap, K-Means effectively differentiates between benign and attack traffic.

### 4.2.2 MODEL EVALUATION AND RESULTS

#### 4.2.2.1 Classification Metrics

```
KMeans Classification Report:
              precision    recall  f1-score   support

           0       0.86      0.95      0.90    453718
           1       0.64      0.36      0.46    111273

    accuracy                           0.83    564991
   macro avg       0.75      0.66      0.68    564991
weighted avg       0.82      0.83      0.82    564991
```

*Fig 4.2.2.1: K-Means Classification Report (Unsupervised Model)*

The K-Means model achieves 0.83 accuracy, performing well on benign traffic (high recall and F1-score) but poorly on attack detection (low recall and F1-score). This shows that while it captures dominant benign patterns, it struggles with complex attack behaviours.

#### 4.2.2.2 Confusion Matrix Analysis

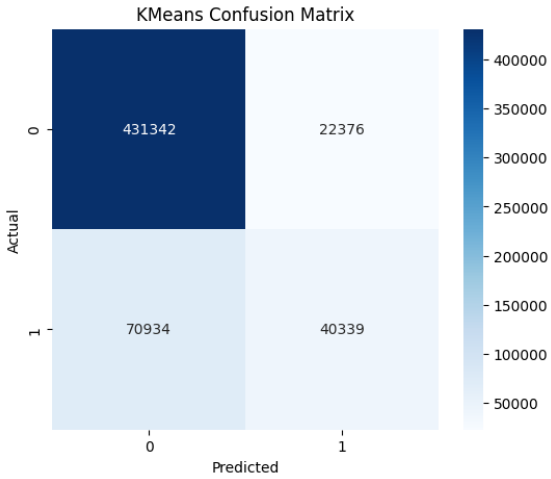


*Fig 4.2.2.2: K-Means Confusion Matrix (Unsupervised Model)*

The confusion matrix shows strong detection of benign traffic (high true negatives) but many attacks are misclassified as benign (high false negatives). This indicates that clustering alone struggles to capture complex attack patterns, reducing detection effectiveness.

#### 4.2.2.3 ROC Curve Analysis

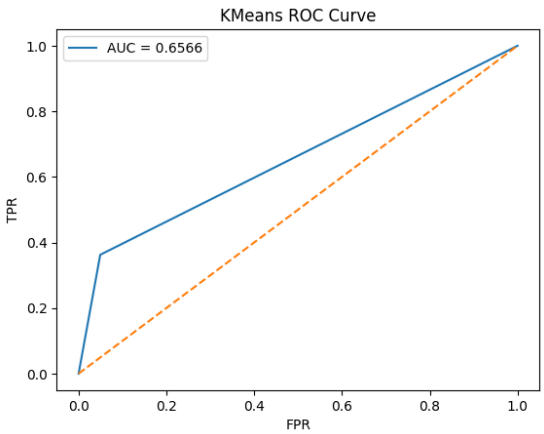


*Fig 2.2.3: K-means ROC Curve (Unsupervised Model)*

The ROC analysis shows an AUC score of 0.6566, indicating that the model performs better than random guessing (AUC = 0.5) but has limited discriminative capability. This further confirms that K-Means, as an unsupervised approach, is not sufficiently effective as a standalone classifier for intrusion detection.

#### 4.2.2.4 Justification for Using K-Means

K-Means is limited as a standalone classifier since it relies on distance-based grouping without labels, struggles with diverse attack patterns, and is affected by class imbalance. However, it is valuable for feature engineering, generating cluster-based features. K-Means clustering captures underlying traffic patterns, and its outputs (clusters and distances) are used as features to better distinguish between benign and malicious traffic, improving final model performance.

### 4.2.3 Conclusion

K-Means clustering captures underlying traffic patterns, and its outputs (clusters and distances) are used as features to better distinguish between benign and malicious traffic, improving final model performance.

## 4.3 MODEL 3 : SUPERVISED LEARNING APPROACH (XGBOOST)

### 4.3.1 SYSTEM OVERVIEW

The final model adopts a hybrid approach by integrating outputs from the supervised Random Forest model and features derived from the unsupervised K-Means clustering process. This fused dataset is used to train the final classification model using XGBoost. XGBoost is selected due to its strong performance on tabular data, ability to model complex feature interactions, and built-in regularization mechanisms that help prevent overfitting. The final model follows a stacked hybrid pipeline:

Raw Features + Random Forest Outputs + Clustering Features → XGBoost Classifier

#### 4.3.1.1 Integration of features from Model 1 & Model 2 into final Data Set

| | Destination Port | Flow Duration | Total Fwd Packets | Total Backward Packets | Total Length of Fwd Packets | Total Length of Bwd Packets | Fwd Packet Length Max | Fwd Packet Length Min | Fwd Packet Length Mean | Fwd Packet Length Std | ... | Active Min | Idle Mean | Idle Std | Idle Max | Idle Min | rf_pred | rf_prob | cluster_dist_0 | cluster_dist_1 | cluster_pred |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **2731287** | 53 | 161 | 2 | 2 | 62 | 94 | 31 | 31 | 31.00000 | 0.000000 | ... | 0 | 0.0 | 0.0 | 0 | 0 | 0 | 0.000000 | 1.963610 | 9.808168 | 0 |
| **2803437** | 443 | 3 | 2 | 0 | 12 | 0 | 6 | 6 | 6.00000 | 0.000000 | ... | 0 | 0.0 | 0.0 | 0 | 0 | 0 | 0.042055 | 4.393625 | 10.294720 | 0 |
| **1920623** | 80 | 5259727 | 4 | 4 | 437 | 839 | 437 | 0 | 109.25000 | 218.500000 | ... | 0 | 0.0 | 0.0 | 0 | 0 | 0 | 0.000000 | 2.802709 | 9.286312 | 0 |
| **644010** | 443 | 3 | 2 | 0 | 12 | 0 | 6 | 6 | 6.00000 | 0.000000 | ... | 0 | 0.0 | 0.0 | 0 | 0 | 0 | 0.042055 | 4.393625 | 10.294720 | 0 |
| **1818676** | 443 | 191849 | 32 | 49 | 871 | 64661 | 389 | 0 | 27.21875 | 83.719476 | ... | 0 | 0.0 | 0.0 | 0 | 0 | 0 | 0.000000 | 6.607804 | 9.663292 | 0 |

*Fig 4.3.1.1: Integration of Features*

The final dataset for Model 3 is constructed by integrating the outputs from both the supervised model (Model 1) and the unsupervised model (Model 2) with the original feature set. Specifically, predictions from the Random Forest model are incorporated as additional features, including the predicted class label **(rf_pred)** and the associated probability score **(rf_prob).**

In parallel, the clustering-based outputs from the K-Means model are also included, comprising the cluster assignment **(cluster_pred)** and the distances of each data point from the cluster centroids **(cluster_dist_0 and cluster_dist_1).** This feature fusion captures both predictive signals and underlying data patterns, providing enriched input for the XGBoost model and improving overall performance.

#### 4.3.1.2 Dataset preparation

The final dataset integrates supervised (rf_pred, rf_prob) and unsupervised (cluster_pred, cluster_dist_0, cluster_dist_1) features to retain both predictive and structural information. Feature selection is performed using Random Forest, selecting the top 20 important features, along with the mandatory hybrid features.The dataset is then split into training and testing sets (80:20) using stratified sampling to maintain class distribution and ensure reliable evaluation.

#### 4.3.1.3 Hyperparameter tuning

XGBoost hyperparameters were optimized using Grid Search with 3-fold cross-validation, evaluating 16 combinations (48 model fits). The best configuration was: learning_rate = 0.1, max_depth = 6,

n_estimators = 200, and subsample = 0.8. This setup balances model complexity and generalization, improving learning capacity, convergence, and robustness.

### 4.3.2 EVALUATION AND ANALYSIS

#### 4.3.2.1 Classification report

| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| Benign (0) | 1.00 | 1.00 | 1.00 | 90,744 |
| Attack (1) | 1.00 | 1.00 | 1.00 | 22,255 |

*Fig 4.3.2.1: Classification Report of XGBoost Model*

The XGBoost model achieved perfect performance with an accuracy, precision, recall, and F1-score of 1.00 for both benign and attack classes. The model correctly classifies all network traffic instances, with minimal false positives or false negatives, indicating excellent predictive capability.

#### 4.3.2.2 Confusion Matrix Analysis

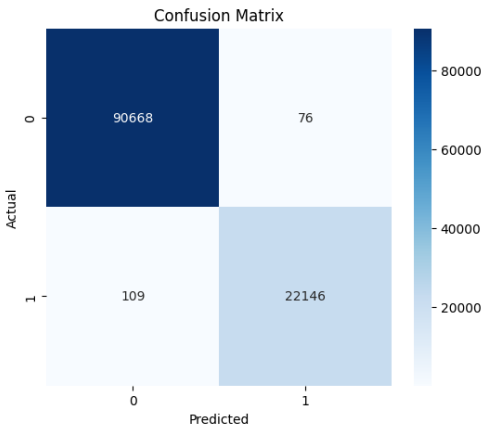


*Fig 4.3.2.2: Confusion Matrix (XGBoost Model)*

The confusion matrix indicates that the XGBoost model achieves highly accurate predictions, with very low false positives (76) and false negatives (109). It correctly classifies the vast majority of benign and attack instances, reflecting strong class separation. This ensures minimal false alarms while effectively detecting malicious traffic, highlighting the model's reliability for intrusion detection.

#### 4.3.2.3 ROC Curve Analysis

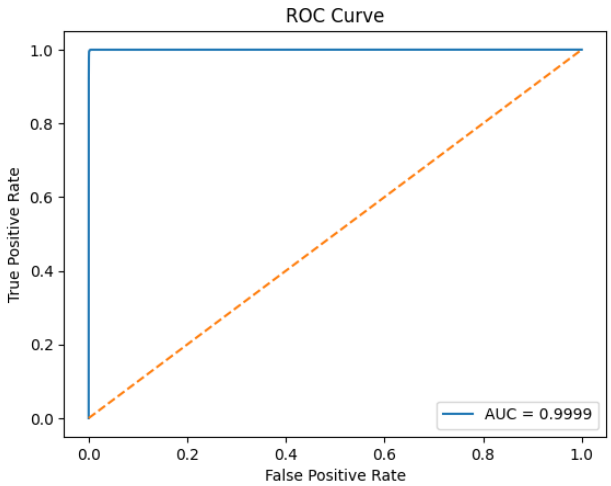


*Fig 4.3.2.3: ROC Curve (XGBoost Model)*

The ROC curve for the XGBoost model demonstrates near-perfect performance, with an AUC score of 0.9999. The curve closely approaches the top-left corner, indicating excellent discriminative ability. This confirms that the model can effectively distinguish between benign and attack traffic with very high accuracy.

#### 4.3.2.4 Learning Curve Analysis

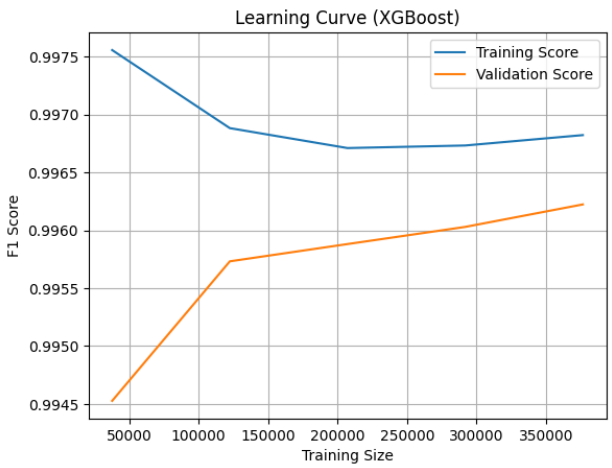


*Fig 4.3.2.4: Learning Curve (XGBoost)*

The learning curve for the XGBoost model shows consistently high training (~0.997) and validation (~0.996) scores with a minimal gap between them. This indicates that the model neither overfits nor underfits and generalizes well to unseen data. Additionally, the steady convergence of the curves suggests that performance improves as the dataset size increases.

#### 4.3.2.5 Feature Importance Analysis

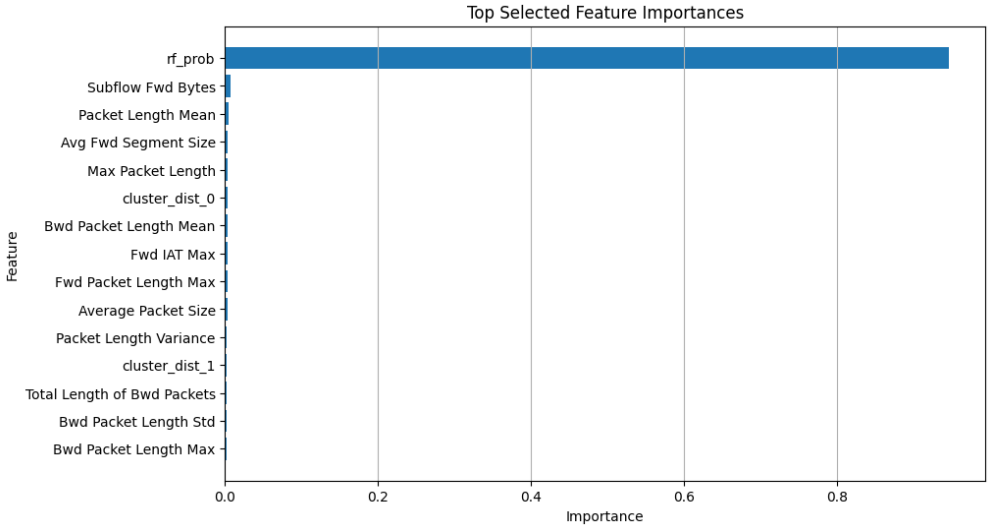


*Fig 4.3.2.5: Top Selected Feature (XGBoost)*

The XGBoost model shows rf_prob (prediction by model 1 ) as the most influential feature, indicating strong predictive power from Random Forest outputs.Clustering features contribute as supportive signals, refining predictions. Overall, this confirms the effectiveness of the hybrid approach.

### 4.3.3 IMPORTANT CONSIDERATION: NEAR-PERFECT PERFORMANCE

While the XGBoost model demonstrates near-perfect performance, these results should be interpreted with caution. The high accuracy may be driven by strong feature engineering (such as *rf_prob* and clustering features), inherently separable patterns in the data, and the ensemble effect of combining multiple models.

## 5. CONCLUSION

The hybrid model demonstrates highly accurate intrusion detection by integrating Random Forest, K-Means clustering, and XGBoost into a unified framework. By combining predictive power with structural data insights, the model achieves strong performance and robustness. This approach highlights the effectiveness of ensemble and hybrid techniques for complex cybersecurity problems, making it a promising solution for real-world network intrusion detection systems.

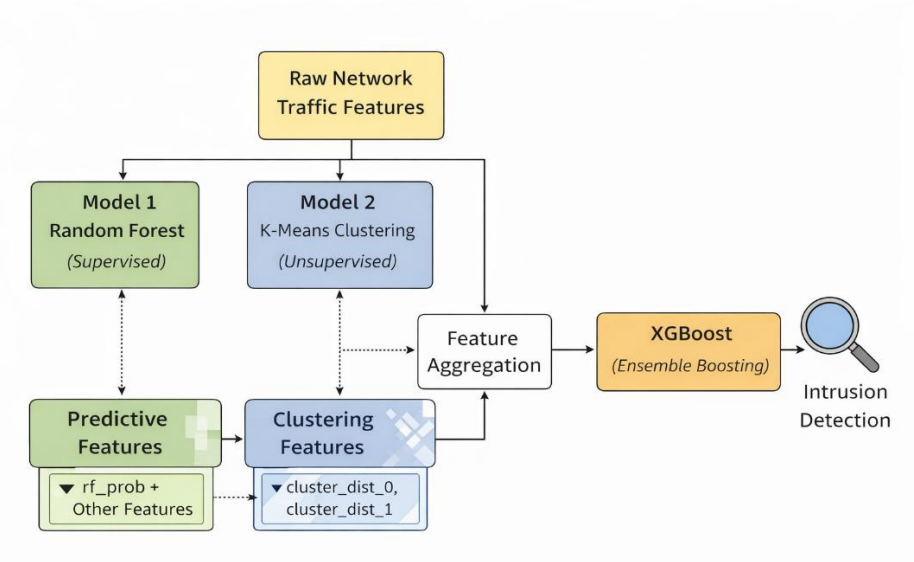


*Fig 5: Schematic Diagram of Hybrid IDS*

## 6. LIMITATIONS AND FUTURE SCOPE

### 6.1 Limitations

The model carries a risk of data leakage due to multi-model feature integration and relies heavily on derived features, making it sensitive to upstream errors. It also has limited interpretability and high computational cost, which can impact real-world deployment. Additionally, performance may vary under changing data distributions and for rare attack classes.

### 6.2 Future Scope

Future work includes building a fully integrated pipeline, exploring advanced or deep learning models, and enabling real-time detection. Enhancing feature engineering, improving model explainability, and extending to multi-class classification will further strengthen the system.

## 7. REFERENCES

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